

A Counterexample to Conjecture 00000007965: Isospectral Inequivalent Binary Codes of Length 6

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Abstract

The second claim of Conjecture 00000007965 asserts that the minimal example of isospectral (same weight enumerator) but non-permutation-equivalent linear codes is the unique pair at $(n, k) = (9, 4)$. We exhibit two binary $[6, 3]$ codes with weight enumerator $1 + 3z^2 + 3z^4 + z^6$ that are not permutation equivalent, so a smaller example exists. We also exhibit two different such pairs of $[9, 4]$ codes, so the pair at $(9, 4)$ is not unique either. Both refutations are checked in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000007965 reads:

Definition: The spectral rigidity of the MacWilliams identity for codes: the isospectral class (the set of equivalence classes of linear codes with the same weight enumerator $A(z)$) under the Krawtchouk transform. Conjecture: The maximal size of an isospectral class at length n and dimension k is exponential $2^{\Omega(n)}$, attained by decomposable codes (direct-sum structure); the minimal example of isospectral but non-permutation-equivalent codes is the unique pair at $(n, k) = (9, 4)$; and the complete spectral rigidity criterion (isospectral implies equivalent) holds in the weight at least 4 part of self-dual codes, uniquely determined by the coefficient classification of modular forms of theta series.

The Chinese version in `conjectures/00000007965.md` states the same three claims.

2 Reading of the statement

The conjecture is a conjunction of three claims. We refute the second one, so the conjecture is false whatever the first and the third claim mean.

Codes. We work with binary linear codes, the setting of the MacWilliams identity with the binary Krawtchouk transform. The statement names no field, and a claim about linear codes in general includes the binary ones. A *binary linear* $[n, k]$ code is a k -dimensional subspace $C \leq \mathbb{F}_2^n$. The Hamming weight $\text{wt}(x)$ of $x \in \mathbb{F}_2^n$ is the number of nonzero coordinates, and $\text{supp}(x) = \{i : x_i \neq 0\}$.

Definition 1. Let $C, D \leq \mathbb{F}_2^n$.

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1. The *weight distribution* of C is $(A_0(C), \dots, A_n(C))$ with $A_w(C) = \#\{c \in C : \text{wt}(c) = w\}$, and its *weight enumerator* is $A_C(z) = \sum_{w=0}^n A_w(C) z^w$.
2. C and D are *isospectral* if $A_C(z) = A_D(z)$.
3. C and D are *permutation equivalent* if there is a permutation σ of $\{1, \dots, n\}$ with $D = \{(c_{\sigma(1)}, \dots, c_{\sigma(n)}) : c \in C\}$.

An *example* is a pair (C, D) of isospectral codes that are not permutation equivalent. Isospectral codes have the same number 2^k of codewords, so both codes of an example have the same parameters (n, k) .

Minimality. The statement does not say in which order $(9, 4)$ is minimal. Every natural choice implies the following weakest form:

$$\text{every example satisfies } n \geq 9 \text{ or } k \geq 4. \quad (\text{M})$$

Indeed, (M) follows if minimality is by length ($n \geq 9$), by dimension ($k \geq 4$), in the componentwise order (an example with $n < 9$ and $k < 4$ would lie strictly below $(9, 4)$), lexicographically in either order, or by $n + k$ ($n + k \geq 13$ rules out $n \leq 8, k \leq 3$).

Uniqueness. “The unique pair at $(9, 4)$ ” means: any two examples of $[9, 4]$ codes are the same pair up to permutation equivalence and order. That is, if (C, D) and (C', D') are examples with $n = 9$ and $k = 4$, then $C \sim C'$ and $D \sim D'$, or $C \sim D'$ and $D \sim C'$, where $\sim$ is permutation equivalence.

3 Results

Theorem 2. Let $A, B \leq \mathbb{F}_2^6$ be the codes with generator matrices

$$G_A = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}, \quad G_B = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

Both are $[6, 3]$ codes with weight enumerator $1 + 3z^2 + 3z^4 + z^6$, and they are not permutation equivalent. Hence (M) fails.

Theorem 3. Let $R = \{000, 111\}$ and $E = \{000, 100\}$, and let $A^R = A \oplus R$, $B^R = B \oplus R$, $A^E = A \oplus E$, $B^E = B \oplus E$ (concatenation of coordinates). These are $[9, 4]$ codes; (A^R, B^R) and (A^E, B^E) are examples; and A^R is permutation equivalent neither to A^E nor to B^E . Hence the pair at $(9, 4)$ is not unique.

Either theorem refutes the second claim, and with it the conjecture.

4 Proofs

Lemma 4. Let $C, D \leq \mathbb{F}_2^n$. Suppose that any two distinct codewords of C of weight 2 have a common support coordinate, while D contains two distinct codewords of weight 2 with disjoint supports. Then C and D are not permutation equivalent. Moreover, permutation equivalent codes have the same weight distribution.

Proof. Suppose $D = \{c \circ \sigma : c \in C\}$. The map $c \mapsto c \circ \sigma$ is a bijection $C \rightarrow D$ with $\text{wt}(c \circ \sigma) = \text{wt}(c)$; this proves the second statement. Let $b_1, b_2 \in D$ be distinct of weight 2 with disjoint supports, and put $a_j = b_j \circ \sigma^{-1} \in C$. Then $a_1 \neq a_2$, both have weight 2, and $\text{supp}(a_j) = \sigma(\text{supp}(b_j))$, so their supports are disjoint. This contradicts the hypothesis on C . $\square$

Proof of Theorem 2. The message $x = (x_1, x_2, x_3)$ is encoded as $xG_A = (x_1, x_1, x_1, x_2, x_3, x_1 + x_2 + x_3)$ and as $xG_B = (x_1, x_1, x_2, x_2, x_3, x_3)$. Both maps are injective, so both codes have dimension 3. Their codewords are

x	000	100	010	001	110	101	011	111
xG_A	000000	111001	000101	000011	111100	111010	000110	111111
xG_B	000000	110000	001100	000011	111100	110011	001111	111111

with weights 0, 4, 2, 2, 4, 4, 2, 6 for A and 0, 2, 2, 2, 4, 4, 4, 6 for B . So $A_C(z) = 1 + 3z^2 + 3z^4 + z^6$ for $C = A$ and for $C = B$.

The weight-2 codewords of A are 000101, 000011, 000110, with supports $\{4, 6\}, \{5, 6\}, \{4, 5\}$; any two of them meet. The weight-2 codewords of B are 110000, 001100, 000011, with pairwise disjoint supports. By Lemma 4, A and B are not permutation equivalent. Thus (A, B) is an example with $n = 6 < 9$ and $k = 3 < 4$, contradicting (M). $\square$

Proof of Theorem 3. For codes $C \leq \mathbb{F}_2^n$ and $F \leq \mathbb{F}_2^m$, the codewords of $C \oplus F$ are the concatenations (c, f) with $\text{wt}(c, f) = \text{wt}(c) + \text{wt}(f)$, so $C \oplus F$ has dimension $\dim C + \dim F$ and $A_{C \oplus F}(z) = A_C(z) A_F(z)$. Hence A^R, B^R, A^E, B^E are $[9, 4]$ codes with

$$\begin{aligned} A_{AR}(z) &= A_{BR}(z) = (1 + 3z^2 + 3z^4 + z^6)(1 + z^3), \\ A_{AE}(z) &= A_{BE}(z) = (1 + 3z^2 + 3z^4 + z^6)(1 + z). \end{aligned}$$

A codeword (c, f) has weight 2 only if $\text{wt}(c) = 2$ and $f = 0$, because A and B have no codewords of weight 1, and the nonzero words of R and E have weights 3 and 1. So for $F \in \{R, E\}$ the weight-2 codewords of $A \oplus F$ (resp. $B \oplus F$) are those of A (resp. B) followed by 000. Their supports pairwise meet in $A \oplus F$ and are pairwise disjoint in $B \oplus F$. By Lemma 4, $A \oplus F$ and $B \oplus F$ are not permutation equivalent, so (A^R, B^R) and (A^E, B^E) are examples.

Finally, A^R has no codeword of weight 1, while A^E and B^E each have exactly one (the word 000000100). By the second part of Lemma 4, A^R is permutation equivalent neither to A^E nor to B^E . So neither alternative in the uniqueness claim holds. $\square$

5 Formalization

The Lean 4 project in `lean/` (file `lean/Submission/Basic.lean`) formalizes the definitions above and proves both theorems against Mathlib.

- `Code n` is `Submodule (ZMod 2) (Fin n -> ZMod 2)`. Hamming weight is Mathlib's `hammingNorm`; dimension is `Module.finrank`.
- `weightDist C w` is the number of codewords of weight w , and `weightEnumerator C` is the polynomial $\sum_{w=0}^n A_w(C) z^w \in \mathbb{N}[z]$. `Isospectral C D` is equality of weight enumerators.
- `PermEquiv C D` is $\exists \sigma, \forall x, x \in C \leftrightarrow x \circ \sigma \in D$, that is, $D = \{c \circ \sigma : c \in C\}$.
- `MinimalAtNineFour` is (M); `UniqueAtNineFour` is the uniqueness statement; `ConjectureClause2` is their conjunction.
- The codes are the row spaces of the generator matrices, defined as `LinearMap.range G.vecMulLinear`: `codeA`, `codeB` are A, B , and `codeA1`, `codeB1`, `codeA2`, `codeB2` are A^R, B^R, A^E, B^E . Injectivity of the encoders, the weight distributions and the weight-2 support facts are checked by `decide` over all messages.
- General lemmas: `weightDist_range`, `finrank_range`, `hammingNorm_comp_perm`, `weightDist_eq_of_permEquiv`, `not_permEquiv_of_disjoint` (Lemma 4).

- Main results: `isospectralInequivalent_codeA_codeB`, `not_minimalAtNineFour`, `not_uniqueAtNineFour`, `conjecture_00000007965_false` ($\neg \text{ConjectureClause2}$), and `conjecture_00000007965_false'`, which states that for any propositions P, Q standing for the first and third claims, $\neg(P \wedge \text{ConjectureClause2} \wedge Q)$.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms: `#print axioms` reports exactly `propext`, `Classical.choice` and `Quot.sound` for `conjecture_00000007965_false` and `not_uniqueAtNineFour`.

6 Remarks

Remark 5. Length 6 is the smallest possible, and (A, B) is the only example up to equivalence with $n \leq 6$. The script `search.py` (Python 3, standard library) enumerates every subspace of $\mathbb{F}_2^n$ for $n \leq 6$, classifies them up to coordinate permutation, and groups the classes by weight distribution. For $n \leq 5$ no two inequivalent codes share a weight distribution. For $n = 6$ exactly one such group occurs, at $k = 3$ with distribution $(1, 0, 3, 0, 3, 0, 1)$, and it consists of two classes. They are the classes of A and B : in the printed representatives, the weight-2 supports pairwise meet in the first and are pairwise disjoint in the second. This remark is not used in the refutation.

Remark 6. Taking $A \oplus F$ and $B \oplus F$ for other codes F gives further examples at $(9, 4)$ and at larger parameters. We do not address the first and third claims of the conjecture.